

Phase Transitions in Price-Weighted Routing Games: Learning Dynamics at the Knife Edge

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CI-enforced theory tests in the public ‘orcap’ repository.*

Abstract

AI inference marketplaces route trillions of tokens per month through a softmax over provider prices with a *published* exponent (selection probability $\propto p^{-a}$; deployed $a = 2$). We study independent Q-learning across this one-parameter family of pricing games. The family has an exact phase structure: symmetric equilibrium price $p^* = ca(n-1)/(a(n-1)-n)$ diverges on the critical line $a(n-1) = n$, an entry-proof markup floor $1/a$ holds everywhere, and the deployed parameter places two-provider markets exactly at criticality. Our object of study is the **interaction of learning dynamics with this phase structure**. Empirically (all runs seeded, manifested, and gated behind a pre-registered validation of the environment against a live market panel — including an untargeted moment, demand elasticity -0.65 ± 0.35 simulated vs -0.78 measured): (1) learned prices are a smooth, strictly decreasing function of a that tracks the equilibrium correspondence away from criticality but **regularizes the singularity** — near the critical line the profit gradient toward higher prices vanishes at rate governed by $h(p) \rightarrow 1$ and ε -greedy Q cannot climb it, so learners undershoot divergent equilibria and overshoot interior Nash ($\Delta = 0.31$ at $a = 2.5$); (2) a measured history-dependent steering penalty (weight $\times 0.17$ on recent price cutters, 7-epoch memory) restructures the learning problem — we prove it deters cutting for any discount below $\delta^\dagger = 0.9895$ and observe it flip learners from bottom-of-book pricing to the menu ceiling in 4/4 calibrated markets (+81% on the flow-dominant quote where the measured conditional binds); (3) learners systematically fail to discover the high-frequency micro-adjustment strategy present in the real market (a $4.9\times$ share tax under the deployed steering makes it dominated), and converge to exact price ties at the market’s focal anchor without any salience assumption — the real market’s two most distinctive regularities are attractors of the learning dynamics. We further evaluate candidate mechanisms with learning agents: a thickness-adaptive exponent removing criticality, and a verified-quality weighting whose closed-form threshold $b^* = 0.63$ separates learned quality-shading from learned quality provision. Our results suggest that deployed routing rules are a natural laboratory where learning theory, mechanism design, and live-market validation meet—and that near singular mechanism parameters, learned play is predictable precisely where equilibrium analysis is not.

1 Introduction

Learning in games meets its cleanest applied test where a mechanism is (i) deployed at scale, (ii) documented exactly, and (iii) instrumented. AI inference routing is such a mechanism: a softmax over provider prices with a published exponent decides, request by request, who serves each call to a shared open-weight model. The exponent is an inverse temperature; the induced game is logit Bertrand — the very testbed of the algorithmic-collusion literature (Calvano et al., 2020) — except that here the demand system is not a modeling choice: it is the product documentation.

This paper treats the routing exponent as the control parameter of a game family and asks how independent learners behave across it, especially at its singular points. The family has a critical line $a(n-1) = n$ where the symmetric equilibrium correspondence diverges to the menu ceiling; the deployed parameter sits on it for duopolies. Away from statistical mechanics analogies, this is a concrete learning question: what does ε -greedy tabular Q do when the stage game’s best-response dynamics point toward an equilibrium at infinity along a vanishing gradient?

Contributions. (1) A complete characterization of the game family’s phase structure with CI-verified closed forms, including an entry-proof elasticity bound (share elasticity $\leq a$) and a measured-elasticity finite version of the critical divergence (duopoly $p^* = 41c$ at the marketplace’s measured outside elasticity). (2) A calibrated environment with a pre-registered validation gate; behavioral species fitted from a live panel; counterfactuals code-locked behind the gate.

(3) The learning phenomenology of the phase structure: smooth regularization of the singularity; overshoot (supra-Nash, Δ up to 0.47) in the disciplined phase; undershoot at criticality; mechanistic explanation via the vanishing gradient of h . (4) Steering as reward shaping: a measured penalty parameter, a deterrence theorem with a patience boundary, and its empirical effect on learned play in calibrated markets. (5) Mechanism evaluation with learners: adaptive temperature; quality weighting with a closed-form learnability threshold.

The router’s softmax makes “inverse temperature” literal, not metaphorical: allocation is a Gibbs measure with energy $a \log p$. The phase-transition language below is the exact behavior of the equilibrium correspondence, not an analogy.

2 The game family

n providers; prices $p_i \in (0, \bar{p}]$; router selects i with probability $s_i = p_i^{-a} / \sum_j p_j^{-a}$; profit $\pi_i = s_i(p_i - c_i)$. Lemma 1: own-price share elasticity $-a(1 - s_i) \in (-a, 0)$. Lemma 2 (single crossing): best responses are unique — root of $h(p) = a(1 - s)(p - c)/p = 1$ or the ceiling. Theorem 1: interior symmetric equilibrium $p^* = c a(n - 1)/(a(n - 1) - n)$ iff $a(n - 1) > n$, ceiling otherwise; Lerner floor $1/a$ in any interior equilibrium (any n , any outside option); with measured outside elasticity ε : $p/(p - c) = a(n - 1)/n + |\varepsilon|/n$ (duopoly at deployed parameters: 41c). Theorem 2 (steering): penalty $\times \theta$ for M epochs after a cut $\Rightarrow$ optimal deviation $c + \sqrt{c^2 + \theta/W}$; deterrence iff $\theta \leq \theta^*$ ($\in [0.81, 1]$ at calibrated configurations, measured $\theta = 0.17$); patience boundary $\delta^\dagger = 0.9895$; perpetual-cutter share tax $\theta/(\theta + n - 1)$ vs $1/n$. Proofs in Appendix A; 13 closed forms CI-tested.

3 Environment and validation

(As in the companion papers; summarized.) Provider agents are behavioral species fitted on the live panel’s train window (adopters / static undercutters / active undercutters / premium; split-sample validated); demand replays the panel’s AR(1)-diurnal process; costs are identified bands from a capital-structure registry and the GPU spot book. The pre-registered gate requires ten moments to match (distance ≤ 0.04 , achieved 0.019, 20 seeds, deterministic replication), with the evidential weight on two emergent moments: untargeted flow elasticity (-0.65 ± 0.35 vs -0.78) and out-of-sample adopter-atom persistence (0.83 vs 0.834). Learning experiments use expected-allocation rewards (exact under non-binding capacity; the request-level kernel supplies sampling noise in the validation runs) and Calvano hyperparameters ($\alpha = 0.15$, $\gamma = 0.95$, exponentially decaying ε), with convergence defined by greedy-policy stability.

4 Learning across the phase diagram

4.1 The dial. Converged mean prices ($n = 3$, 8 seeds/point): uniform 1.53; $a = 1$: 1.30; $a = 2$: 0.94; $a = 4$: 0.60; $a \geq 8$: grid floor. Learned price is smooth and strictly decreasing in a — the mechanism’s comparative statics survive bounded rationality everywhere.

4.2 At criticality. For $n = 3$ the critical point is $a = 1.5$. There, theory says the symmetric equilibrium is the ceiling (1.6); learners reach $1.098 \pm \text{small}$. Mechanism: near criticality $d\pi/dp \propto 1 - h(p)$ with $h \rightarrow 1$, so the drift toward the ceiling is arbitrarily weak relative to exploration noise; tabular Q with decaying ε performs a random walk on a nearly flat profit landscape and settles far below the divergent equilibrium. Conversely at $a = 2.5$ (interior Nash 0.5) learners hold 0.877 — supra-Nash by $\Delta = 0.31$: in the disciplined phase the landscape has curvature and the classic Calvano reward-punishment channel operates. **Learning friction regularizes the equilibrium correspondence: undershoot where it diverges, overshoot where it is interior.** We view this as the paper’s most transferable observation: mechanism analysis that stops at the equilibrium correspondence mispredicts learned play in *both directions*, in a sign pattern that a vanishing-gradient argument predicts.

4.3 What is not learned. Substituting a learner into each behavioral slot (8/8 seeds unanimous): micro-adjustment (~ 1 reprice/day) never emerges — under the deployed steering it carries a $4.9\times$ share tax and is dominated (Theorem 2(iv)); the learner instead parks at a rigid below-anchor price. With no static undercutter present, the learner converges to the anchor price *exactly*: endogenous tie formation. The real market’s tie atom (45% of same-model pairs, $3.4\times$ its

mechanical null) and its bifurcation into rigid-vs-micro-adjusting technologies are thus both attractors of learning under the deployed mechanism — an out-of-sample style of validation we find more convincing than moment matching: the environment reproduces *qualitative regularities it was never told about.*

4.4 Steering as reward shaping. The measured penalty is a history-dependent, asymmetric transformation of the reward: cuts move the agent into a taxed state for M epochs. Effect on learned play: stylized world — learner jumps 0.72 $\rightarrow$ ceiling (8/8 seeds), market +18%. Calibrated markets — ceiling 4/4 under the broad rule; under the strictly-measured cheapest-only conditional, the effect concentrates at the book’s bottom: +81% on the flow-dominant quote where the learner is cheapest, escape otherwise (both variants reported; they bracket the unobserved rule). The penalty converts “undercut” from the greedy action into a trap state; a $\gamma = 0.95$ learner ($\delta^M \approx 0.70 \ll \delta^\dagger = 0.9895$) can never profit from entering it.

5 Mechanism evaluation with learning agents

5.1 Objective frontier under cost heterogeneity. Two calibrated cost types (owned-capacity 0.10×2 ; spot-dependent 0.50×3). Theory FOCs and learner runs [E-MECH1] agree on the ordering: welfare $\uparrow$ in a (better cost-sorting), ad-valorem platform revenue $\downarrow$ in a, spot-type viability $\downarrow 0$ (the resilience cost of sharp routing). The deployed ($a = 2$, ad-valorem fee) pair is revenue-serving, not welfare-serving; a thickness-adaptive $a(n) = n/(\ell(n-1))$ pins markups and removes the critical line. **5.2 Quality learnability threshold.** Binary quality action; shading saves $\Delta = 0.08$ at hidden damage $d = 0.2$. Weight $w = q^b p^{-a}$: all-high-quality is an equilibrium iff $b \geq b^* = 0.63$ (closed form). E-MECH2 (5 seeds, 600k epochs): learned hi-quality share rises 0.27 ($b = 0$, the deployed quality-blind rule) $\rightarrow$ 0.40 ($b = 0.5$) $\rightarrow$ 0.67 ($b = 1$) $\rightarrow$ 0.53 ($b = 2$). The direction matches the theory — quality weighting shifts learned play toward quality provision, with the largest jump crossing b^* — but the bifurcation is noisy rather than sharp: the quality margin ($\Delta = 0.08$) is small relative to price-learning noise in the doubled action space, and at $b = 2$ quality-weight and price competition interact. We report this as directional confirmation of the closed-form threshold, not a sharp learnability boundary. The q signal exists operationally (our deployed daily evaluation probes: graded benchmarks and greedy-output hashes per provider).

6 Related work

Algorithmic pricing and collusion: Calvano et al.; Klein (sequential moves); Asker et al. (learning protocols); Abada-Lambin (exploration- driven pseudo-collusion); Hansen-Misra-Pai (misspecified bandits); Deng-Schiffer-Bichler (deep RL nearer Nash). Our departure: the demand system is deployed and documented, the environment is validation-gated against a live market, and the phase-structure/learning interaction is the object. Steering: Johnson-Rhodes-Wildenbeest (anti-collusive steering design) — we analyze the measured inverse. Mechanism/learning interfaces: reward shaping, potential-based equivalences; the cut-penalty is a naturally occurring non-potential shaping with measured parameters. Marketplace empirics: Demirer et al. (OpenRouter). Softmax/temperature in games: logit QRE (McKelvey-Palfrey) — our learners’ smooth dial echoes QRE comparative statics, but the regularization sign pattern at criticality is, to our knowledge, undocumented.

7 Limitations

Tabular Q with one hyperparameter suite (Calvano’s), chosen for comparability; a systematic hyperparameter/algorithm sweep (optimism, UCB, PPO) is the natural next step and prior evidence suggests deep RL lands nearer Nash — which would sharpen, not reverse, §4.2’s contrast. Expected-allocation rewards; symmetric-equilibrium theory; steering θ measured for one buyer tier; short calibration panel with a registered 30-day re-estimation that automatically reopens claims on sign flips.

8 Reproducibility statement

Public repository with: frozen calibration bundles (train/holdout recorded), pre-registration documents with dated addenda, per-run manifests (commit, bundle hash, seed tree, source fingerprints), 46 tests including 13 enforcing every closed form in §2 against continuum numerics, and the live-capture workflows that generated the panel.

9 Appendix A: proofs. Appendix B: run tables (E-SIM1–4b, E-MECH1–2).